Andhra University College of Engineering

Computer Science and Engineering | Course Code: CS302

Database Management Systems - Comprehensive Syllabus Lecture Notes

Syllabus Focus: CS302 Full Syllabus Comprehensive Lecture Notes Units 1-5

Summary: Complete syllabus lecture notes covering Database Management Systems. Includes detaile

I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

Unit 1: Relational Data Model & Formal Languages

" Syllabus Scope: Database system architectures, Entity-Relationship ER modeling, Relational algebra oper

Unit 2: Relational Query Language SQL

" Syllabus Scope: DDL, DML, DCL commands, Nested subqueries, Correlated subqueries, Aggregate functions,

CS302: Database Management Systems - Advanced Topics & Assessment

Academic Track: Computer Science and Engineering - Units 3, 4, 5 & Examination Matrix

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II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

Unit 3: Relational Database Design & Normalization

" Functional dependencies, Armstrong axioms, Attribute closure, 1NF, 2NF, 3NF, Boyce-Codd Normal Form BCNF, 4N

Unit 4: Transaction Processing & Concurrency Control

" ACID properties, Serializability, Conflict vs View serializability, Two-Phase Locking 2PL, Strict 2PL, Times

Unit 5: Storage, Indexing & Query Processing

" RAID levels, B-Trees, B+ Trees node splitting and deletion, Static and Dynamic hashing, Cost-based query exe

III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

1. Given a relation RA, B, C, D, E with functional dependencies, compute candidate keys and decompose R i
2. Prove whether a given concurrent transaction schedule is conflict serializable by constructing its pre
3. Construct a B+ Tree index of order 4 for a sequence of 10 search keys, illustrating all node splitting
4. Explain the Two-Phase Locking 2PL protocol and prove how it guarantees conflict serializable schedules

IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* Raghu Ramakrishnan, Johannes Gehrke - Database Management Systems McGraw-Hill 3rd Ed
* Abraham Silberschatz, Henry F. Korth - Database System Concepts McGraw-Hill 7th Ed

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